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U-substitution.

$$\int \overset{\text{Outside}}{\underset{\text{function}}{f}} \left(\overset{\text{inside function}}{g} \left(\overset{\text{Derivative of}}{\underset{\text{inside function}}{x}} \right) \right) \overset{\text{Derivative of}}{\underset{\text{inside function}}{g'(x)}} dx = F(g(x)) + C$$

Recognizing the $f(g(x))g'(x)$ form.

$$\begin{aligned} & \int x(x^2 + 1)^2 dx \\ & u = x^2 + 1 \\ & du = 2x dx \\ & \frac{du}{2} = \frac{2x dx}{2} \Rightarrow x dx = \frac{du}{2} \\ & \int x(x^2 + 1)^2 dx \Rightarrow \int \frac{1}{2} \cdot (u)^2 du \\ & = \frac{1}{2} \int u^2 du \\ & = \frac{1}{2} \cdot \frac{u^3}{3} + C \\ & \frac{u^3}{6} + C \end{aligned}$$

$$\int \sqrt{2x-1} dx$$

$$u = 2x - 1 \Rightarrow du = 2dx$$

$$\int \sqrt{u} \cdot \frac{du}{2}$$

$$\frac{1}{2} \int u^{1/2} du$$

$$\frac{1}{2} \cdot \frac{2}{3} u^{3/2} + C$$

$$\therefore \int \sqrt{2x-1} dx = \frac{1}{3} \cdot (2x-1)^{3/2} + C$$

1/22/2024

Change of Variables.

$$\int_0^1 x(x^2+1)^3 dx$$

$$\text{Let } u = x^2 + 1$$

$$du = 2x dx$$

$$\frac{1}{2} du = x dx$$

$x = 0$	$x = 1$
$u = 1$	$u = 2$

$$\int_0^1 x(x^2+1)^3 dx \Rightarrow \frac{1}{2} \int_1^2 u^3 du$$

$$= \frac{1}{2} \cdot \frac{u^4}{4} \Big|_1^2 = \frac{15}{8}$$

Integration of Trigonometric Functions.

$$\begin{aligned} & \int \sec x dx \\ & \int \frac{\sec x (\sec x + \tan x)}{(\sec x + \tan x)} dx \\ & \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx \\ & \ln |\sec x + \tan x| + C \end{aligned}$$

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$$\begin{aligned} & \int (1 + \cot^2 x) dx \\ & \int \csc^2 x dx = \cot x + C \end{aligned}$$

#2

$\int \frac{(\ln x)^3}{x} dx$	$u = \ln x$
	$du = \frac{1}{x} dx$
$\int u^3 du$	
$\frac{1}{4} u^4 + C$	
$\frac{1}{4} (\ln x)^4 + C$	

#10 Diagnostic Test.

$$\int \frac{\sqrt[7]{3 - \ln z^6}}{z} dz$$

Logarithmic Properties

$$\log_b b = 1$$

$$\log_b a^c = c \log_b a$$

$$a^{\log_a b} = b$$

$$\log a - \log b = \log \frac{a}{b}$$

$$\log a + \log b = \log ab$$

$$\int \frac{\sqrt[7]{3 - \ln z^6}}{z} dz$$

$$\int \frac{\sqrt[7]{3 - 6 \ln z}}{z} dz$$

$$\begin{aligned} u &= 3 - 6 \ln z \\ du &= -6 \cdot \frac{1}{z} dz \end{aligned}$$

$$\int \sqrt[7]{3 - 6 \ln z} \cdot \frac{-6 dz}{-6z}$$

$$-\frac{1}{6} \int u^{1/7} du = -\frac{1}{6} \cdot \frac{u^{\frac{1}{7}+1}}{\frac{1}{7}+1} + C$$

$$-\frac{7}{48} (3 - 6 \ln z)^{\frac{8}{7}} + C$$

1/23/2024

Differentiate implicitly the equation $x = \ln(x + y + 1)$ to find the slope of the tangent line at any given point (x,y)

$$\begin{aligned}\frac{d}{dx}(x) &= \frac{d}{dx}(\ln(x+y+1)) \\ 1 &= \frac{1}{x+y+1} \cdot \frac{d}{dx}(x+y+1) \\ 1 &= \frac{1}{x+y+1} \cdot 1 + y' + 0 \\ 1 + y' &= x + y + 1 \\ y' &= x + y \\ \text{Random Slope of } (-1, e^{-1}) \\ \hline y' &= -1 + \frac{1}{e}\end{aligned}$$

Basic Integrals and Derivatives

$\frac{d}{dx}(\sin u) = \cos u \, du$	$\int \cos u \, du = \sin u + C$
$\frac{d}{dx}(\cos u) = -\sin u \, du$	$\int \sin u \, du = -\cos u + C$
$\frac{d}{dx}(\tan u) = \sec^2 u \, du$	$\int \sec^2 u \, du = \tan u + C$
$\frac{d}{dx}(\sec u) = \sec u \tan u \, du$	$\int \sec u \tan u \, du = \sec u + C$
$\frac{d}{dc}(\cot u) = -\csc^2 u \, du$	$\int \csc^2 u \, du = -\cot u + C$
$\frac{d}{dc}(\csc u) = -\csc u \cot u \, du$	$\int \csc u \cot u \, du = -\csc u + C$

5.5 Derivative of exponential functions.

$\frac{d}{dx}(2^x)$	$\frac{d}{dx}(e^u) = e^u du$ $a^{\log_a b} = b$ Remember that if $a^u = x$ then $\log_a x = u$ Logarithm is another word for exponent!!
$\therefore \frac{d}{dx}(2^x) = \frac{d}{dx}[e^{\ln 2^x}]$	Remember that $\ln(x) = \log_e(x)$
$\therefore \frac{d}{dx}(2^x) = \frac{d}{dx}[e^{\ln 2^x}] = e^{\ln 2^x} \frac{d}{dx}(\ln 2^x)$	
$= e^{\ln 2^x} \frac{d}{dx}[x \cdot \ln 2]$	
$= 2^x \ln 2$	
$\therefore \frac{d}{dx}(a^u) = a^u (\ln a)$	

5.5#4

Find the derivative of $y = 5^{-2x}$

$$\frac{d}{dx}[a^u] = a^u (\ln a) du$$

$$\frac{d}{dx}[5^{-2x}] = 5^{-2x} (\ln 5)(-2)$$

5.5#5

Find the derivative of the function $f(x) = x5^x$

$$f(x) = x5^x$$

$$f'(x) = \frac{d}{dx}(x) \cdot 5^x + x \cdot \frac{d}{dx}(5^x)$$

$$= 1 \cdot 5^x + x \cdot 5^x \cdot \ln 5 \cdot \frac{d}{dx}(x)$$

$$= 5^x + x \cdot 5^x \ln 5$$

$$= 5^x (1 + x \ln 5)$$

Integration of $\int a^u du$

$\int a^{\omega} d\omega$	Remember $a^u = e^{\ln a^u}$
$\int e^{\omega} d\omega$	$\int e^{\ln a^u} du$
$e^{\omega} + C$	$\omega = \ln a^u$
	$\omega = u \ln a$
	$\frac{d\omega}{du} = \ln a$
	$d\omega = \ln a du$
	$\frac{d\omega}{\ln a} = du$
$\therefore \int e^{\omega} d\omega$	$\int e^{\ln a^u} du \Rightarrow \int e^{\omega} \cdot \frac{d\omega}{\ln a}$
$= \frac{1}{\ln a} \int e^{\omega} d\omega$	
$= \frac{1}{\ln a} e^{\omega} + C$	
$\therefore \int a^u du = \frac{1}{\ln a} a^u + C$	

5.5#9

Find the integral of $\int 7^{-x} dx$

$$\int 7^{-x} dx$$

$$\begin{array}{l} a = 7 \quad u = -x \\ \frac{du}{dx} = -1 \quad du = -dx \end{array}$$

$$\begin{aligned} -\int 7^u du &= -\int e^{\ln 7^u} du \\ &= -\int e^w dw \end{aligned}$$

$$w = \ln 7^u \quad w = u(\ln 7) \quad \frac{dw}{du} = \ln 7 \quad dw = \ln 7 du$$

$$\begin{aligned} &= -\frac{1}{\ln 7} \int e^{\ln 7^u} du \\ &= -\frac{1}{\ln 7} \cdot e^{\ln 7^u} + C \\ &= -\frac{1}{\ln 7} \cdot 7^u + C \end{aligned}$$

#23 Exam Review

$$\int_1^2 6x^2 4x^3 dx$$

$$\begin{array}{l} u = x^3 \quad du = 3x^2 2x \\ 2du = 6x^2 2x \\ x = 1, u = 1 \quad x = 2, u = 8 \end{array}$$

$$\begin{aligned} \int_1^8 4^u du &= 2 \cdot \frac{1}{\ln 4} \cdot 4^u \Big|_1^8 \\ &= 2 \left[\frac{1}{\ln 4} \right] (4^8 - 4) \end{aligned}$$

$$\begin{aligned} &= \frac{2}{\ln 4} (65536 - 4) \\ &= \frac{2}{\ln 4} (65532) \\ &= \frac{2}{\ln 2^2} (65532) \\ &= \frac{1}{\ln 2} (65532) \end{aligned}$$

Exam Review #24

$\int_1^2 \frac{6 \ln x}{x} dx$	$\frac{1}{\ln 6} [6^{\ln 2} - 6^0]$
$u = \ln x \quad du = \frac{1}{x} dx$	$\frac{1}{\ln 6} [6^{\ln 2} - 1]$
$\int_0^{\ln 2} 6^u du$	$\frac{6^{\ln 2} - 1}{\ln 6}$
$\left. \frac{1}{\ln 6} 6^u \right _0^{\ln 2}$	

5.7 Inverse Trigonometric Functions: Differentiation.

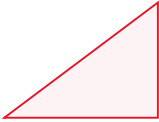
Key Concepts.

- For a function to have a derivative, it needs to be one-to-one.
- Trigonometric functions need to have their domain limited for them to be one-to-one.

If given a function such as $y = \arcsin u$ and asked to find the derivative, we proceed as follows.

$y = \arcsin u$	$\cos y y' = du$
$\sin y = u$	$y' = \frac{du}{\cos y}$
$\frac{d}{dx}(\sin y) = \frac{d}{dx}(u)$	

If given $y = \arcsin e^{x^2}$ what is $y' = ?$



$$r = e^{x^2}$$

$q =$

$\sin y = e^{x^2} = \frac{e^{x^2}}{1}$	$\cos y \cdot y' = 2xe^{x^2}$
$\frac{d}{dx}(\sin y) = \frac{d}{dx}[e^{x^2}]$	$y' = \frac{2xe^{x^2}}{\cos y}$
$\cos y \cdot y' = e^{x^2} \cdot 2x$	$y' = \frac{2xe^{x^2}}{\sqrt{1 - e^{2x^2}}}$

Since

$$\frac{d}{dx}(\arcsin u) = \frac{u'}{\sqrt{1-u^2}}$$

Then we can say that

$$\int \frac{du}{\sqrt{1-u^2}} = \arcsin u + C$$

5.8#4

Find the integral

$$\int \frac{6x+7}{\sqrt{1-x^2}} dx$$

We split it into two in order to fit the arcsin

$$\int \frac{6x dx}{\sqrt{1-x^2}} + \int \frac{7}{\sqrt{1-x^2}} dx$$

For $\int \frac{6x dx}{\sqrt{1-x^2}}$

$$u = 1-x^2 \quad du = -2x$$

$$-3 \int \frac{du}{\sqrt{u}} = -3 \int u^{-\frac{1}{2}} du$$

For $\int \frac{7}{\sqrt{1-x^2}} dx$

$$= 7 \int \frac{1}{\sqrt{1-x^2}} dx$$

Combining the two yields

$$C - 6\sqrt{1-x^2} + 7 \sin^{-1} x$$

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Exam Review #23

$$\int_1^2 6x^2 4^{x^3} dx$$

$$u = x^3 \quad du = 3x^2 dx$$

$$x=1, u=1$$

$$x=2, u=8 \quad dx = \frac{du}{3x^2}$$

$$2 \int_1^8 4^u du$$

$$2 \left[\frac{1}{\ln 4} 4^u \right]_1^8$$

$$\frac{2}{\ln 2^2} \left[4^u \right]_1^8$$

$$\frac{1}{\ln 2} \left[4^8 - 4 \right]$$

Exam I Review#24

$\int_1^2 \frac{6^{\ln x}}{x} dx$	$u = \ln x$	$du = \frac{1}{x} dx = \frac{dx}{x}$
$\int_1^2 6^{\ln x} \frac{dx}{x}$ $\int 6^u du = \frac{1}{\ln 6} \cdot 6^u = \frac{1}{\ln 6} [6^{\ln x}]_1^2$ $= \frac{1}{\ln 6} [6^{\ln 2} - 6^{\ln 1}]$		

5.8 Inverse Trigonometric Functions: Integration

It is important to note that,

➤	$\frac{d}{dx} [\arcsin u] = -\frac{d}{dx} [\arccos u]$	
	$\frac{d}{dx} [\arctan u] = -\frac{d}{dx} [\operatorname{arccot} u]$	
	$\frac{d}{dx} [\operatorname{arcsec} u] = -\frac{d}{dx} [\operatorname{arccsc} u]$	

➤ In order to master a formula, learning how to derive it a sure way.

➤ The composition of a function and its inverses simply gives a variable as shown below

$(f \circ g)(x) = x$ $y = x^2 - 4 \quad [-\infty, 0]$ For instance $x = y^2 - 4 \quad [-4, \infty]$ $\pm\sqrt{x+4} = y$ $g(x) = f^{-1}(x) = (-\sqrt{x+4})$	$f(g(x)) = (g(x))^2 - 4$ $= (-\sqrt{x+4})^2 - 4$ $= x + 4 - 4 = x$
--	--

$$\frac{d}{dx} [\arcsin u]$$

$\sin y = \sin(\arcsin u)$ $\sin y = u$ $\frac{d}{dx} [\sin y] = \frac{d}{dx} u$ $\cos y \cdot y' = du$	$y' = \frac{du}{\cos y}$ $\sin y = u$ $\cos y = \frac{u}{1}$
--	--

Find derivative of $y = \arcsin \ln(x^2)$

$y' =$ $\sin y = \sin \left[\arcsin \ln(x^2) \right]$ $\sin y = \ln x^2$ $\frac{d}{dx} [\sin y] = \frac{d}{dx} [\ln x^2]$ $\cos y \cdot y' = \frac{2x}{x^2} = \frac{2}{x}$	$y' = \frac{2}{x \cos y}$ $\text{but } \cos y = \frac{\sqrt{1 - (\ln x^2)^2}}{1}$ $y' = \frac{2}{x \sqrt{1 - (\ln x^2)^2}}$
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Find the Derivative of arcsine (u)

$\operatorname{arccsc} u$ $\csc y = \csc(\operatorname{arccsc} u)$ $\frac{d}{dx} [\csc y] = \frac{d}{dx} (u)$ $-\cot y \csc y \cdot y' = u'$ $y' = -\frac{u'}{\cot y \csc y}$	$= \frac{-u'}{\frac{\sqrt{u^2 - 1}}{1} \frac{u}{1}}$ $= -\frac{u'}{u \sqrt{u^2 - 1}}$
Triange with; $\text{opp} = 1, \text{hyp} = u, \text{adj} = \sqrt{u^2 - 1}$	

Thursday, February 1, 2024

5.8 Inverse Trig Functions: Integration.

$\frac{d}{dx} \left(\arctan \frac{u}{a} \right)$ $\tan y = \tan \left(\arctan \frac{u}{a} \right)$ $\frac{d}{dx} [\tan y] = \frac{d}{dx} \left[\frac{u}{a} \right]$ Triangle: Opp=u, adj=a, hyp=$\sqrt{a^2 + u^2}$ $\sec^2 y \cdot y' = \frac{1}{a} \cdot du$ $y' = \frac{\frac{1}{a} du}{\sec^2 y} = \frac{du}{a \sec^2 y}$	$= \frac{du}{a \left(\frac{\sqrt{a^2 + u^2}}{a} \right)^2}$ $= \frac{adu}{a^2 + u^2}$ $\frac{d}{dx} \left(\arctan \frac{u}{a} \right) = \frac{adu}{a^2 + u^2}$ $\Rightarrow \int \frac{adu}{a^2 + u^2} = \arctan \frac{u}{a} + C$ $a \int \frac{du}{a^2 + u^2} = \arctan \frac{u}{a} + C$ $\int \frac{du}{a^2 + u^2} = \frac{1}{a} \cdot \arctan \frac{u}{a} + C$
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5.8#1

Find the indefinite integral.

$\int \frac{16}{1+64x^2} dx$ $16 \int \frac{dx}{1+64x^2} \left\{ \begin{array}{l} a=1, u=8x \\ du=8dx \\ dx=\frac{8}{du} \end{array} \right\} \left\{ \frac{1}{a} \arctan \frac{u}{a} + C \right\}$ $= \frac{16}{8} \cdot \frac{1}{1} \arctan \frac{8x}{1} + C$ $= 2 \arctan 8x + C$	<u>Proof</u> $\frac{d}{dx} [16 \arctan 8x + C]$ $y = \arctan 8x + C$ $\tan y =$ Triangle: Opp=8x, adj=1 hyp=$\sqrt{64x^2 + 1}$ $y = \arctan 8x + C$ $\tan y = 8x$ $\frac{d}{dx} [\tan y] = \frac{d}{dx} [8x]$ $\sec^2 y \cdot y' = 8$ $y' = (2) \frac{8}{\sec^2 y} = (2) \frac{8}{\left(\frac{\sqrt{64x^2 + 1}}{1} \right)^2}$ $= \frac{16}{64x^2 + 1}$
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5.8#3

Find the indefinite integral.

$\int \frac{e^{2x}}{16 + e^{4x}} dx$	<table border="1" style="margin: auto;"> <tr> <td style="padding: 5px;">$a = 4$</td> <td style="padding: 5px;">$u = e^{2x}$</td> </tr> <tr> <td style="padding: 5px;">$\frac{du}{dx} = 2e^{2x}$</td> <td style="padding: 5px;">$du = 2e^{2x} dx$</td> </tr> </table>	$a = 4$	$u = e^{2x}$	$\frac{du}{dx} = 2e^{2x}$	$du = 2e^{2x} dx$
$a = 4$	$u = e^{2x}$				
$\frac{du}{dx} = 2e^{2x}$	$du = 2e^{2x} dx$				
$\frac{1}{2} \int \frac{du}{a^2 + u^2} = \frac{1}{2} \cdot \frac{1}{a} \cdot \arctan \frac{u}{a} + C$ $= \frac{1}{2} \cdot \frac{1}{4} \arctan \frac{1}{4} e^{2x} + C$					

Derivation of the derivative of cosecant u.

$y = \operatorname{arccsc} \frac{u}{a}$ $\csc y = \frac{u}{a}$ Triangle opp=a, hyp=u adj=$\sqrt{u^2 - a^2}$ $\frac{d}{dx}(\csc y) = \frac{d}{dx}\left(\frac{u}{a}\right)$ $-\csc y \cot y \cdot y' = \frac{1}{a} \cdot du$ $y' = \frac{du}{a(-\csc y \cot y)}$	$= -\frac{du}{-a \cdot \frac{u}{a} \cdot \frac{\sqrt{u^2 - a^2}}{a}}$ $= -\frac{du}{\frac{u\sqrt{u^2 - a^2}}{a}}$ $\frac{d}{dx}\left(\operatorname{arccsc} \frac{u}{a}\right) = -\frac{adu}{u\sqrt{u^2 - a^2}}$ $\int \frac{-adu}{u\sqrt{u^2 - a^2}} = \operatorname{arccsc} \frac{u}{a}$ $-\frac{1}{a} \int \frac{-adu}{u\sqrt{u^2 - a^2}} = -\frac{1}{a} \operatorname{arccsc} \frac{u}{a}$ $\int \frac{du}{u\sqrt{u^2 - a^2}} = -\frac{1}{a} \operatorname{arccsc} \frac{ u }{a} + C$
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5.8#5

$\int_1^3 \frac{1}{x\sqrt{16x^2 - 4}} dx$ $a = 2 \quad u = 4x \quad du = 4dx$ $\frac{4}{4} \int_1^3 \frac{4}{x4\sqrt{16x^2 - 4}} dx = \frac{1}{2} \operatorname{arccsc} \frac{ 4x }{2} \Big _1^3$	$= \frac{1}{2} \left[\operatorname{arccsc} \frac{4(3)}{2} - \operatorname{arccsc} \frac{4(1)}{2} \right]$ $= \frac{1}{2} (\operatorname{arccsc} 6 - \operatorname{arccsc} 2)$ $= \frac{1}{2} \left(\operatorname{arccsc}(6) - \frac{\pi}{2} \right)$
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Hyperbolic Functions.

Definitions of the Hyperbolic Functions.

$\sinh x = \frac{e^x - e^{-x}}{2}$	$\operatorname{csch} x = \frac{1}{\sinh x}, \quad x \neq 0$
$\cosh x = \frac{e^x + e^{-x}}{2}$	$\operatorname{sech} x = \frac{1}{\cosh x}$
$\tanh x = \frac{\sinh x}{\cosh x}$	$\operatorname{coth} x = \frac{1}{\tanh x}, x \neq 0$

Monday, February 5, 2024

Derivative of $\sinh x$

$y = \sinh x$		
$y' = ?$		
$y = \frac{e^x - e^{-x}}{2}$		
$y = \frac{1}{2}(e^x - e^{-x})$		
$y' = \frac{1}{2}[e^x(1) - e^{-x}(-1)]$		
$y' = \frac{1}{2}[e^x + e^{-x}]$		
$\therefore \int \cosh x = \sinh x + C$		

Proof of $\cosh^2 x - \sinh^2 x = 1$

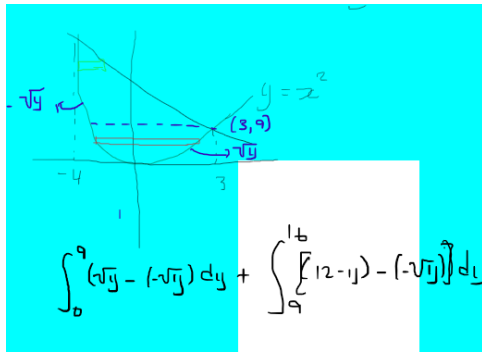
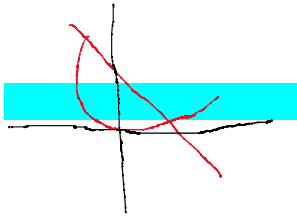
$\sinh^2 x - \cosh^2 x = -1$		
$\sinh x = \frac{e^x - e^{-x}}{2}$		
$\cosh x = \frac{e^x + e^{-x}}{2}$		
$\frac{1}{4}(e^{2x} - 2e^x e^{-x} + e^{-2x}) - \frac{1}{4}(e^{2x} + 2e^x e^{-x} + e^{-2x}) \stackrel{?}{=} -1$		
$e^{2x} - 2e^x e^{-x} + e^{-2x} - (e^{2x} + 2e^x e^{-x} + e^{-2x}) = -4$		
$e^{2x} - 2 + e^{-2x} - e^{2x} - 2 - e^{-2x} = -4$		
$-4 = -4$		
Which was an error but also able to prove that		
$\cosh^2 x - \sinh^2 x = 1$		

5.9#4

Find the derivative of $\sinh 6x$		
$y' = \cosh 6x \cdot 6$		
$y' = 6 \cosh 6x$		

7.1#3

Consider the following. $Y=x^2$, $y=12-x$



Find limits $x^2 = 12 - x$ $(x + 4)(x - 3) = 0$ $x = -4, 3$ $\int_{-4}^3 12 - x - (x^2) dx$ $12x - \frac{1}{2}x^2 - \frac{1}{3}x^3 \Big _{-4}^3$ $12(3) - \frac{1}{2}(3)^2 - \frac{1}{3}(3)^3 - \left[12(-4) - \frac{1}{2}(-4)^2 - \frac{1}{3}(-4)^3 \right]$ $36 - \frac{9}{2} - 9 - \left(-48 - 8 + \frac{64}{3} \right)$	$\int_0^9 2\sqrt{y} dy + \int_9^{16} (12 - y + \sqrt{y}) dy$ $2 \int_0^9 y^{1/2} dy$ $2 \cdot \frac{2}{3} y^{3/2} \Big _0^9 + 12 - \frac{1}{2} y^2 + \frac{2}{3} y^{3/2} \Big _9^{16}$ $12(16) - \frac{1}{2}(16)^2 + \frac{2}{3}(4^2)^{3/2} - \left[12(9) - \frac{1}{2}(81) + \frac{2}{3}(3^2)^{3/2} \right]$ $12(16) - \frac{1}{2} \cdot 16 \cdot 16 + \frac{2}{3} \cdot 64 - 12(9) + \frac{81}{2} - 18$ $12(16 - 9) + \frac{81}{2} - 146$	
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Tuesday, February 6, 2024

5.5#6

$h(x) = \log_3 \frac{x\sqrt{x-6}}{3}$ $= \log_3 x\sqrt{x-6}$ $= \log_3 x\sqrt{x-6} - 1$ $= \log_3 x + \log_3 \sqrt{x-6} - 1$ $= \log_3 x + \log_3 (x-6)^{\frac{1}{2}} - 1$ $h(x) = \log_3 x + \frac{1}{2} \log_3 (x-6) - 1$ $= \frac{\ln x}{\ln 3} + \frac{1}{2} \left[\frac{\ln(x-6)}{\ln 3} \right] - 1$	$h'(x) = \frac{1}{\ln 3} \cdot \frac{1}{x} + \frac{1}{\ln 9} \cdot \frac{1}{x-6}$ $h'(x) = \frac{1}{\ln 3^3} + \frac{1}{\ln 9^{(x-6)}}$	
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5.5#7

$\ln y = \ln x^{\frac{3}{x}}$ $\ln y = \frac{3}{x} \ln x$ $\frac{1}{y} \cdot y' = 3 \cdot \frac{d}{dx} \left[\frac{1}{x} \cdot \ln x \right]$ $\frac{y'}{y} = 3 \left[-\frac{1}{x^2} \cdot \ln x + \frac{1}{x} \cdot \frac{1}{x} \right]$	$\frac{y'}{Y} = 3 \left[-\frac{1}{x^2} \ln x + \frac{1}{x^2} \right]$ $y' = y \left\{ \frac{3}{x^2} [1 - \ln x] \right\}$ $y' = x^{3/x} \left\{ \frac{3}{x^2} [1 - \ln x] \right\}$	
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5.5#12

$\int_{-2}^2 36^{\frac{1}{2}x} dx$ $\int a^u du = \frac{1}{\ln a} a^u + C$		
$u = \frac{1}{2}x \quad \frac{du}{dx} = \frac{1}{2}$ $du = \frac{1}{2} dx$	if $x=-2, u=-1$ $x=2, u=1$	
$\int_{-1}^1 36^u 2 du$ $2 \int_{-1}^1 36^u du$ $2 \left[\frac{1}{\ln 36} 36^u \right]_{-1}^1$ $\frac{2}{\ln 36} [36^1 - 36^{-1}]$	$\frac{1}{\ln 6} \left(36 - \frac{1}{36} \right)$ $\frac{1}{\ln 6} \left(\frac{36^2 - 1}{36} \right)$ $\frac{36^2 - 1}{36 \ln 6}$ $\frac{36^2 - 1}{\ln 6^{36}}$	

5.7#9

$\int_1^2 \frac{2x-3}{\sqrt{4x-x^2}}$ $\int \frac{2x-3}{\sqrt{4x-x^2}} dx$ $\int \frac{2x}{\sqrt{4x-x^2}} dx - 3 \int \frac{dx}{\sqrt{4x-x^2}}$ $-\int \frac{(-1)2x}{\sqrt{4x-x^2}} dx - 3 \int \frac{dx}{\sqrt{4x-x^2}}$ $-\int \frac{-2x+4-4}{\sqrt{4x-x^2}} dx - 3 \int \frac{dx}{\sqrt{4x-x^2}}$ $-\int \frac{4-2x}{\sqrt{4x-x^2}} dx + 4 \int \frac{dx}{\sqrt{4x-x^2}} - 3 \int \frac{dx}{\sqrt{4x-x^2}}$	$\int \frac{dx}{\sqrt{4x-x^2}} - \int \frac{4-2x}{\sqrt{4x-x^2}} dx$ $\int \frac{dx}{\sqrt{2^2-(x-2)^2}} - \int \frac{4-2x}{\sqrt{4x-x^2}} dx$ $= \arcsin \frac{(x-2)}{a} \Big _1^2 - \int \frac{du}{u}$ $= \arcsin \frac{(x-2)}{a} \Big _1^2 - 2\sqrt{u} \Big _1^2$ $= \arcsin \frac{(x-2)}{a} \Big _1^2 - 2\sqrt{u} \Big _1^2$	$= \sin^{-1}(0) - \arcsin\left(-\frac{1}{2}\right) - 2[2 - \sqrt{3}]$ $= 0 - \left(-\frac{\pi}{6}\right) - 4 + 2\sqrt{3}$ $= \frac{\pi}{6} - 4 + 2\sqrt{3}$
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5.9 Hyperbolic Functions.

Proof of $\cosh^2 x - \sinh^2 x = 1$

$$\left(\frac{e^x + e^{-x}}{2}\right)^2 - \left(\frac{e^x - e^{-x}}{2}\right)^2 = 1$$

$$\frac{e^{2x} + 2 \cdot e^x e^{-x} + e^{-2x}}{4} - \frac{e^{2x} - 2e^x e^{-x} + e^{-2x}}{4} = 1$$

$$\frac{e^{2x} + 2 + e^{-2x} - e^{2x} + 2 - e^{-2x}}{4} = 1$$

$$\frac{4}{4} = 1$$

$$1 = 1$$

Proof of $\frac{d}{dx}(\sinh u) = \cosh u$

$$\frac{d}{dx}(\sinh u) = \frac{d}{dx} \left[\frac{e^u - e^{-u}}{2} \right]$$

$$\frac{1}{2} \cdot \frac{d}{dx} [e^u - e^{-u}]$$

$$= \frac{1}{2} [e^u du - e^{-u} (-1) du]$$

$$= \cosh u$$

Proof of $\sinh x = \ln(x + \sqrt{x^2 + 1})$

$$y = \sinh x = \frac{e^x - e^{-x}}{2}$$

$$x = \sinh y = \frac{e^y - e^{-y}}{2}$$

$$2x = e^y - \frac{1}{e^y}$$

$$2x = \frac{e^{2y} - 1}{e^y}$$

$$2xe^y = e^{2y} - 1$$

$$0 = e^{2y} - 2xe^y - 1$$

$$e^y = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(1) \pm \sqrt{(-2x)^2 - 4(1)(-1)}}{2(1)}$$

$$= e^y = \frac{2x \pm \sqrt{4x^2 + 4}}{2}$$

$$= \frac{2x \pm 2\sqrt{x^2 + 1}}{2}$$

$$\frac{2(x \pm \sqrt{x^2 + 1})}{2} = e^y$$

$$e^y = x + \sqrt{x^2 + 1}$$

$$\ln e^y = \ln(x + \sqrt{x^2 + 1})$$

$$y = \ln(x + \sqrt{x^2 + 1})$$

derivative of $\left[\sinh^{-1} u\right] = \frac{u'}{\sqrt{u^2 + 1}}$ $\frac{d}{dx}\left[\sinh^{-1} u\right] = \frac{d}{dx}\left[\ln\left(u + \sqrt{u^2 + 1}\right)\right]$ $1du + \frac{1}{\sqrt{u^2 + 1}}$ $= \frac{1}{\sqrt{u^2 + 1}}$	NOT FINISHED		
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Wednesday, February 7, 2024

7.2#4

Find the volume of the solids generated by revolving the region bounded by the graphs of the equations about the given lines.

$y = \sqrt{x}$ $y = 0$ $x = 3$ $V = \pi \int_0^3 (\sqrt{x} - 0)^2 dx$ $= \pi \int_0^3 x dx$ $= \pi \left[\frac{x^2}{2} \right]_0^3 = \frac{9\pi}{2}$		
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$\pi \int_0^{\sqrt{3}} (9 - y)^2 - (9 - 3)^2 dx$ $\pi \int_0^{\sqrt{3}} 81 - 18y + y^2 - 9 dx$ $= \frac{144\pi\sqrt{3}}{5}$

7.3 Shell Method.

$\int_0^2$ The volume of the shell is given by $r - \frac{w}{2} = \text{radius of hole}$ $r + \frac{w}{2} = \text{radius of outside cylinder}$ $V = \pi r^2 h$ $V_0 = \pi \left(r + \frac{w}{2} \right)^2 h$ $V_H = \pi \left(r - \frac{w}{2} \right)^2 h$	Volume of Shell $V_0 - V_H$ $\pi \left(r + \frac{w}{2} \right)^2 h - \pi \left(r - \frac{w}{2} \right)^2 h$ $\pi h \left[r^2 + rw + \frac{w^2}{4} - \left(r^2 - rw + \frac{w^2}{4} \right) \right]$ $\pi h \left[r^2 + rw + \frac{w^2}{4} - r^2 + rw - \frac{w^2}{4} \right]$ $= \pi h 2rw$ $\therefore V_{\text{SHELL}} = 2\pi \int_0^2 \overbrace{r(x)h(x)dy}^{\text{Radius Height Thickness}}$	For our example we have $2\pi \int_0^2 y(4 - y^2) dy$ $2\pi \int_0^2 (4y - y^3) dy$ $2\pi \left[4 \cdot \frac{1}{2} y^2 - \frac{1}{4} y^4 \right]_0^2$ $2\pi \left[2 \cdot 2^2 - \frac{1}{4} \cdot 2^4 - \left(2 \cdot 0^2 - \frac{1}{4} \cdot 0 \right) \right]$ $2\pi [8 - 4]$ $= \boxed{8\pi}$
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Monday, February 12, 2024

7.1#7

Consider the following functions. $f(x) = 5 \sin x$ $g(x) = 5 \cos 2x$ $-\frac{\pi}{2} \leq x \leq \frac{\pi}{6}$ Sketch the region bounded by the graphs $\int_{-\frac{\pi}{2}}^{\frac{\pi}{6}} (5 \cos 2x - 5 \sin x) dx$ $5 \int_{-\frac{\pi}{2}}^{\frac{\pi}{6}} (\cos 2x - \sin x) dx = 5 \int_{-\pi}^{\frac{\pi}{3}} \cos u \frac{du}{2}$ $u = 2x$ $du = 2dx$ $\frac{du}{2} = dx$ if $x = -\frac{\pi}{2}, u = 2\left(-\frac{\pi}{2}\right) = -\pi$ if $x = \frac{\pi}{6}, u = 2\left(\frac{\pi}{6}\right) = \frac{\pi}{3}$	$= \frac{5}{2} \int_{-\pi}^{\frac{\pi}{3}} \cos u du$ $\frac{5}{2} \cdot \sin u \Big _{-\pi}^{\frac{\pi}{3}}$ $\frac{5}{2} \left(\sin \frac{\pi}{3} - \sin(-\pi) \right)$ $= \frac{5}{2} \left[\frac{\sqrt{3}}{2} - 0 \right] = \frac{5\sqrt{3}}{4}$	
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Tuesday, February 13, 2024

7.1#7

$$\begin{aligned} & \int_0^2 [(-x+4) - (x^2-9)] dx \\ & \int_{-\pi/2}^{\pi/6} (9 \cos 2x - 9 \sin x) dx \\ & 9 \int_{-\pi/2}^{\pi/6} (\cos 2x - \sin x) dx \\ & \dots \end{aligned}$$

7.1#9

$$\begin{aligned} F(x) &= \int_0^x \left(\frac{1}{2}t + 9 \right) dt \\ &= \left[\frac{1}{4}t^2 + 9t \right]_0^x \\ &= \frac{1}{4}x^2 + 0x - (0+0) \\ &= \frac{1}{4}x^2 + 9x \\ F(6) &= \frac{1}{4}(6)^2 + 9 \cdot 6 \\ &= 9 + 54 \\ &= 63 \end{aligned}$$

7.1#11

$$2m \left[2 \left[\int_0^5 \left(1 - \frac{1}{3} \sqrt{5-x} \right) dx + \int_5^{11/2} (1-0) dx \right] m^2 \right]$$

With Disk Method

Find the volume of the solid generated by revolving the region bounded by the given lines and curves about the x-axis.

$$y = \sqrt{\sin 6x}$$

$$y = 0$$

$$0 \leq x \leq \frac{\pi}{6}$$

After drawing the sketch, with at least two representative rectangles

$$\begin{aligned} V &= \pi \int r^2 h \\ &= \pi \int_0^{\pi/6} (\sqrt{\sin 6x} - 0)^2 dx \\ &= \frac{\pi}{6} \int_0^{\pi/2} (\sin 6x)(6) dx = \frac{\pi}{6} \int_0^{\pi} \sin u du \\ &= \frac{\pi}{3} \end{aligned}$$

With Shell Method

$$V = 2\pi \int \overbrace{p}^{\text{Height of Shell}} \overbrace{r}^{\text{Radius of Shell}} \overbrace{w}^{\text{Thickness of Shell}}$$

$$V = 2\pi \int_0^1 y \left[\frac{1}{6} \arcsin(y^2) - \frac{1}{6} \arcsin(y^2) \right] dy$$

Derivation of the arc length Formular.

$$\begin{aligned} &\sum \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &\lim_{x \rightarrow \infty} \sum_{i=1}^n \sqrt{\Delta x_i^2 + \Delta y_i^2} \cdot \frac{\Delta y}{\Delta x} \\ &\sum_{i=1}^n \sqrt{\Delta x_i^2 + \Delta y_i^2} \frac{\Delta x}{\sqrt{\Delta x^2}} \\ &\sum_{i=1}^n \frac{\sqrt{\Delta x_i^2 + \Delta y_i^2}}{\sqrt{\Delta x^2}} \Delta x \\ &\sum \sqrt{\frac{\Delta x_i^2}{\Delta y_i^2} + \frac{\Delta y_i^2}{\Delta x_i^2}} \\ &\dots \\ &\sum \sqrt{1 - [f'(x)]^2} \Delta x \end{aligned}$$

Derivation of the Surface Area of Revolution.

$$Area = 2\pi \int r L dx = 2\pi \int r \sqrt{1 + [f'(x)]^2} dx$$

Since L is the arclength is w

Wednesday, February 14, 2024

Exam Review #4

$\int_1^x \sqrt{t^2 - 1} dt$ Trig sub let $t = \sec \theta$ $t^2 = \sec^2 \theta$ $t^2 - 1 = \tan^2 \theta$ $\sqrt{\tan^2 \theta} = \tan \theta$ $\frac{dt}{d\theta} = \sec \theta \tan \theta$ $\int \tan \theta \sec \theta \tan \theta d\theta$ $\int \tan^2 \theta \sec \theta d\theta$ $\int (\sec^2 \theta - 1)(\sec \theta) d\theta$ $\int (\sec^3 \theta - \sec \theta) d\theta$	$\int \sec^3 \theta d\theta - \int \sec \theta d\theta$ $\int \sec^3 \theta d\theta = \sec \theta \tan \theta - \int \tan \theta \sec \theta \tan \theta d\theta$ $= \sec \theta \tan \theta - \int \tan^2 \theta \sec \theta d\theta + C$ $\int \sec^2 \theta \sec \theta d\theta$ $\int u dv = uv - \int v du$ $u = \sec \theta \quad dv = \sec^2 \theta d\theta$ $du = \sec \theta \tan \theta \quad v = \int \sec^2 \theta d\theta$ $= \tan \theta$ $2 \int \tan^2 \theta \sec \theta d\theta = \sec \theta \tan \theta + C$ $\int \tan^2 \theta \sec \theta d\theta = \frac{1}{2} \sec \theta \tan \theta - \frac{1}{2} \int \sec \theta d\theta$	$\int_1^x \sqrt{t^2 - 1} dt = \int \tan^2 \theta \sec \theta d\theta$ $= -\frac{1}{2} \int \sec \theta d\theta + \frac{1}{2} \sec \theta \tan \theta$ $= -\frac{1}{2} \ln \sec \theta + \tan \theta + \frac{1}{2} \sec \theta \tan \theta$ $\frac{t}{1} = \sec \theta$ Triangle: Opp = $\sqrt{t^2 - 1}$, adj = 1, hyp = t $= -\frac{1}{2} \ln \left \frac{t}{1} + \frac{\sqrt{t^2 - 1}}{1} \right + \frac{1}{2} t \cdot \sqrt{t^2 - 1}$ $= \frac{1}{2} t \sqrt{t^2 - 1} \dots + \sqrt{t^2 - 1}$ $y = \frac{1}{2} x \sqrt{x^2 - 1} - \frac{1}{2} \ln x + \sqrt{x^2 - 1}$
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7.4#9

Set up the integral.

$y = \frac{1}{2} x^3$ $SA = 2\pi \int r L dx$ $SA = 2\pi \int r L dx$ $= 2\pi \int_0^3 \left(\frac{1}{3} x^3 - 0 \right)$ $= 2\pi \int_0^3 \left(\frac{1}{3} x^3 - 0 \right) \sqrt{1 + (x^2)^2} dx$ $= \frac{2\pi}{3} \int_0^3 x^3 \sqrt{1 + x^4} dx$ $u = 1 + x^4$ $\frac{du}{dx} = 4x^3$	$\frac{2\pi}{3} \int x^3 \sqrt{u} \frac{du}{4x^3}$ $\frac{\pi}{6} \int u^{1/2} du$ $x=0, u=1+0^4 = 1$ $x=3, u=1+3^4 = 82$ $\frac{\pi}{6} [82^{3/2} - 1^{3/2}]$ $\frac{\pi}{6} [82^{3/2} - 1]$	
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7.2#8

Find the volume revolved about the x-axis.

$$y = \cos 2x$$

$$y = 0$$

$$x = 0$$

$$x = \frac{\pi}{4}$$

After sketching the solid of revolution with at least two representative rectangles.

$$V = \pi \int r^2 dx$$

$$= \pi \int_0^{\pi/4} (\cos 2x - 0)^2 dx$$

$$= \pi \int_0^{\pi/4} (\cos 2x - 0)^2 dx$$

$$= \pi \int_0^{\pi/4} \cos^2 2x dx$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A + A) = \cos^2 A - \sin^2 A$$

$$\cos(2A) = \cos^2 A - (1 - \cos^2 A)$$

$$\cos 2A = 2 \cos^2 A - 1$$

$$\cos 2A = 2 \cos^2 A - 1$$

$$\frac{1 + \cos 2A}{2} = \cos^2 A$$

$$= \pi \int_0^{\pi/4} \left(\frac{1 + \cos 4x}{2} \right) dx$$

$$= \pi \int_0^{\pi/4} \frac{1 + \cos 4x}{2} dx$$

$$= \frac{\pi}{2} \int_0^{\pi/4} (1 + \cos 4x) dx$$

$$u = 4x \quad x = 0 \quad u = 0$$

$$du = 4dx \quad x = \frac{\pi}{4},$$

$$\frac{du}{4} = dx$$

$$\frac{\pi}{2} \left[\frac{\pi}{4} - 0 + \frac{1}{4} \sin u \right]_0^{\pi}$$

$$= \frac{\pi^2}{8}$$

Thursday, February 15, 2024

7.2#7

$-x^2 + 2x + 6 = x^2 + 2$ $0 = 2x^2 - 2x - 4$ $0 = 2(x^2 - x - 2)$ $x^2 - x - 2 = 0$ $(x - 2)(x + 1) = 0$ $\pi \int_0^2 \left[(-x^2 + 2x + 6)^2 - (x^2 + 2)^2 \right] dx + \pi \int_2^3 \left[(x^2 + 2 - 0)^2 - (-x^2 + 2x + 6)^2 \right] dx$ $\pi \int_0^2 (-4x^3 - 12x^2 + 24x + 32) dx + \int_2^3 (4x^3 + 12x^2 - 24x - 32) dx$ $\pi \left[-2^4 - 4 \cdot 2^3 + 12 \cdot 2^2 + 32 \cdot 2 - 0 \right] + \pi \left[\frac{3^4}{4} + 4 \cdot \frac{3^3}{3} - 12 \cdot \frac{3^2}{2} - 32 \cdot 3 - \left(\frac{2^4}{4} + 4 \cdot \frac{2^3}{3} - 12 \cdot \frac{2^2}{2} - 32 \cdot 2 \right) \right]$ $\pi \left[-16 - 32 + 48 + 64 \right] + \pi \left[\frac{81 + 108 - 108 - 96}{4} - \frac{16 + 32 - 48 - 64}{4} \right]$ $= 113\pi$		
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7.2#6

<u>Washer Method</u>	<u>Disk Method</u>	
$y = 3 - x$ $x = 3 - y$ $\pi \int_0^2 \left[(3 - 0)^2 - (3 - (3 - y))^2 \right] dy$ $\pi \int_0^2 \left[3^2 - [3 - 3 + y]^2 \right] dy$ $\pi \int_0^2 (9 - y^2) dy$ $\pi \left[9y - \frac{1}{3}y^3 \right]_0^2 = \pi \left[9 \cdot 2 - \frac{1}{3} \cdot 2^3 \right]$ $\pi \left[18 - \frac{8}{3} \right] = \frac{46\pi}{3}$	$2\pi \int_0^1 prw + 2\pi \int_1^3 prw$ $2\pi \int_0^1 (2 - 0)(3 - x) dx + 2\pi \int_1^3 (3 - x - 0)(3 - x) dx$ $2\pi \int_0^1 2(3 - x) dx + 2\pi \int_1^3 (3 - x)^2 dx$ $2\pi \int_0^1 (6 - 2x) dx + 2\pi \int_1^3 (9 - 6x + x^2) dx$ $2\pi \left[6x - x^2 \right]_0^1 + \left[9x - 3x^2 + \frac{1}{3}x^3 \right]_1^3$ $2\pi \left[6 - 1 - 0 \right] + \left(27 - 27 + 9 - \left(9 - 3 + \frac{1}{3} \right) \right)$ $2\pi \left[5 + \left(5 + 2\frac{2}{3} \right) \right]$ $\frac{46\pi}{3}$	

7.4#5

Find the arclength of the graph over the indicated interval.

7.4#9

$SA = 2\pi \int rL$ $= 2\pi \int_a^b r \sqrt{1 + (y')^2} dx$ $= 2\pi \int_0^3 \left(\frac{1}{3}x^3 - 0 \right) \sqrt{1 + (x^2)^2} dx$ $= 2\pi \int_0^3 \left(\frac{1}{3}x^3 \sqrt{1 + x^4} \right) dx$ $= \frac{2\pi}{3} \int_0^3 \left(x^3 \sqrt{1 + x^4} \right) dx$ $u = 1 + x^4$ $\frac{du}{dx} = 4x^3 \quad \frac{du}{4x^3} = dx$ $\frac{2\pi}{3} \int \cancel{x^3} \sqrt{u} \frac{du}{4 \cancel{x^3}}$ $\frac{\pi}{6} \int \sqrt{u} du = \frac{\pi}{6} \int u^{1/2} du$	$x = 0 \Rightarrow u = 1 + 0^4 = 1$ $x = 3 \Rightarrow u = 1 + 81 = 82$ $\frac{\pi}{6} \left[\frac{2}{3} u^{3/2} \right]_1^{82}$ $\frac{\pi}{9} [82^{3/2} - 1^{3/2}]$ $\frac{\pi}{9} (82^{3/2} - 1)$	
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Thursday, February 22, 2024

Exam 1 Revision.

$\int_1^{\sqrt{2}} x 9^{x^2} dx$ $u = x^2$ $\frac{du}{dx} = 2x$ $du = 2x dx$ $\int_1^2 9^u \frac{du}{2}$ $\text{if } x=1, u=1^2 = 1$ $\text{if } x=\sqrt{2}, u = (\sqrt{2})^2$ $\int a^u du = \frac{1}{\ln a} a^u + C$		
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Exam 1 Review.

$y = -\cot^{-1}\left(\frac{8x}{3}\right)$ $y = \operatorname{arccot}\left(\frac{8x}{3}\right)$ $\cot y = \left(\frac{8x}{3}\right)$ Triangle: Opp=3, Adj=8x, hyp=$\sqrt{64x^2 + 9}$ $-\csc^2 y \cdot y' = \frac{8}{3}$ $y' = \frac{\frac{8}{3}}{-\csc^2 y} = \frac{-8}{3 \cdot \frac{64x^2 + 9}{9}}$	$= \frac{-24}{64x^2 + 9}$ $\therefore \frac{24}{64x^2 + 9}$	
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Exam 1 Review

$\int_0^{\frac{1}{3}} \frac{5x dx}{\sqrt{16 - x^4}}$ $5 \int_0^{1/3} \frac{x dx}{\sqrt{16 - x^4}}$ $a^2 = 16 \quad u^2 = x^4$ $a = 4 \quad \boxed{u = x^2}$ $\int \frac{du}{a^2 - u^2} = \sin^{-1} \frac{u}{a} + C$ $\int \frac{du}{u\sqrt{u^2 - a^2}} = \frac{1}{a} \sec^{-1} \frac{ u }{a} + C$ $\int \frac{du}{a^2 + u^2} = \frac{1}{a} \tan^{-1} \frac{u}{a} + C$	$5 \int \frac{du/2}{\sqrt{a^2 - u^2}}$ $\frac{5}{2} \int \frac{du}{\sqrt{a^2 - u^2}} = \frac{5}{2} \arcsin \frac{x^2}{4} + C$ $= \frac{5}{2} \left[\sin^{-1} \frac{x^2}{4} \right]_0^{\frac{1}{3}}$ $\frac{5}{2} \left[\sin^{-1} \frac{1/9}{4} - \sin^{-1} 0 \right]$ $= \frac{5}{2} \sin^{-1} \frac{1}{36}$	
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Exam 1 Review

$\int \frac{dt}{t^2 + 8t + 20}$		
$\int \frac{dt}{t^2 + 8t + 16 + 4}$		
$\int \frac{dt}{(t+4)^2 + 2^2}$		
$= \frac{1}{2} \tan^{-1} \left(\frac{t+4}{2} \right) + C$		

Exam 1 Review

$y = 7^{\cos \theta}$		
$y = a^u$		
$y' = \ln u (a^u) + C$		
$= (\ln 7) 7^{\cos \theta} (-\sin \theta)$		

Exam 1 Question Review.

$\ln y = \ln x^{10 \sin x}$		
$\frac{d}{dx} [\ln y] = \frac{d}{dx} [10 \sin x \ln x]$		
$\frac{1}{y} \cdot y' = 10 \left[\cos x \cdot \ln x + \frac{1}{x} \sin x \right]$		
$y' = 10x^{10 \sin x} \left(\cos x \ln x + \frac{\sin x}{x} \right)$		

Exam 1 Question Review

$\int \cosh^2 \left(4 - \frac{x}{4} \right) dx$	$2\pi \int_0^{\pi/4} \sin x \sqrt{1 + \cos^2 x} dx$	
$\int \cosh^2 u du = -\coth u + C$		
$\int \operatorname{csch}^2 u \cdot (-4 du)$		
$-4 \int \operatorname{csch}^2 u du$		
$-4 \left(-\coth \left(4 - \frac{1}{4} x \right) \right) + C$		
$4 \coth \left(4 - \frac{1}{4} x \right) + C$		

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